20260507_BEL2_Mathematics

Vectors, Matrices & Tensors

FOR LOGISTICS ENGINEERS.

Scalar : one value — demand $= 5 \ \text{parts}/\text{week}$

vector : one index x_i — demand on 3 stores

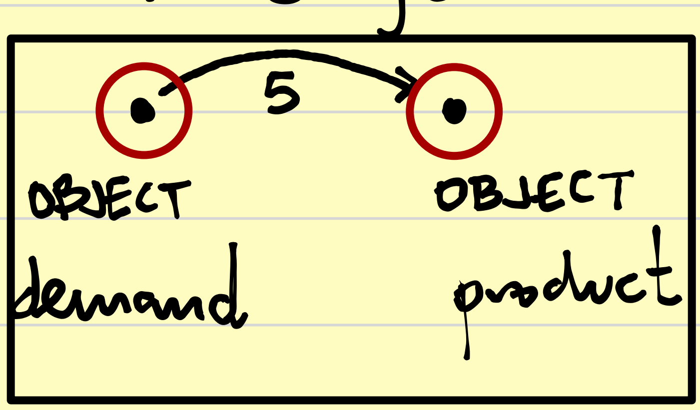
$$\vec{d} = \begin{bmatrix} 120 \\ 80 \\ 100 \end{bmatrix} \ \text{parts}/\text{week}$$

matrix : two indexes a_{ij} — Shipment plan from 2 warehouses to 3 stores

$$A = \begin{array}{c} \underbrace{\begin{bmatrix} 20 & 30 & 10 \\ 40 & 10 & 20 \end{bmatrix}}_{\text{3 stores}} \end{array} \Big] \text{2 WH}$$

tensor : three or more indexes t_{ijk}

SCALAR. Single number



$$d = 5$$

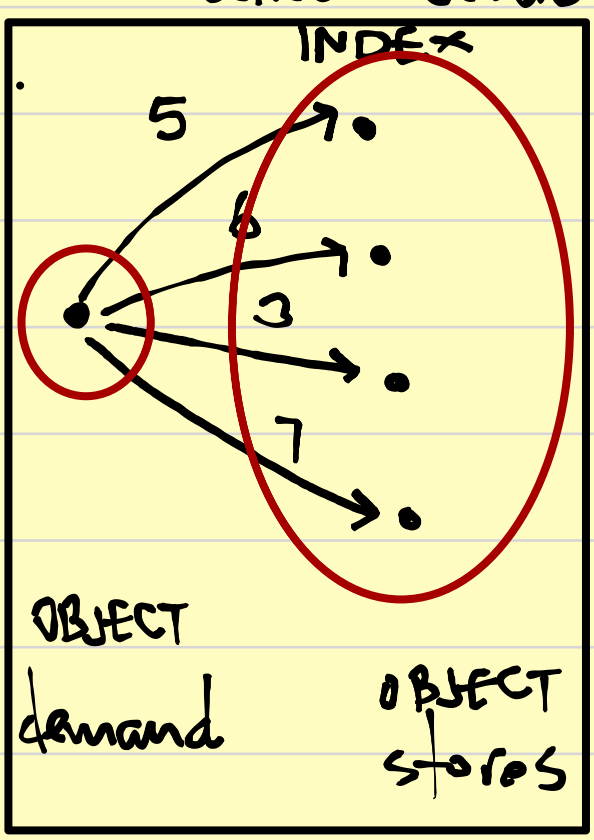
VECTOR. An ordered list of numbers

Column vector: $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$

Row vector: $x^T = [x_1 \; x_2 \cdots x_n] \in \mathbb{R}^n$

x^T. Transposed

A vector collects several related quantities of the same kind.



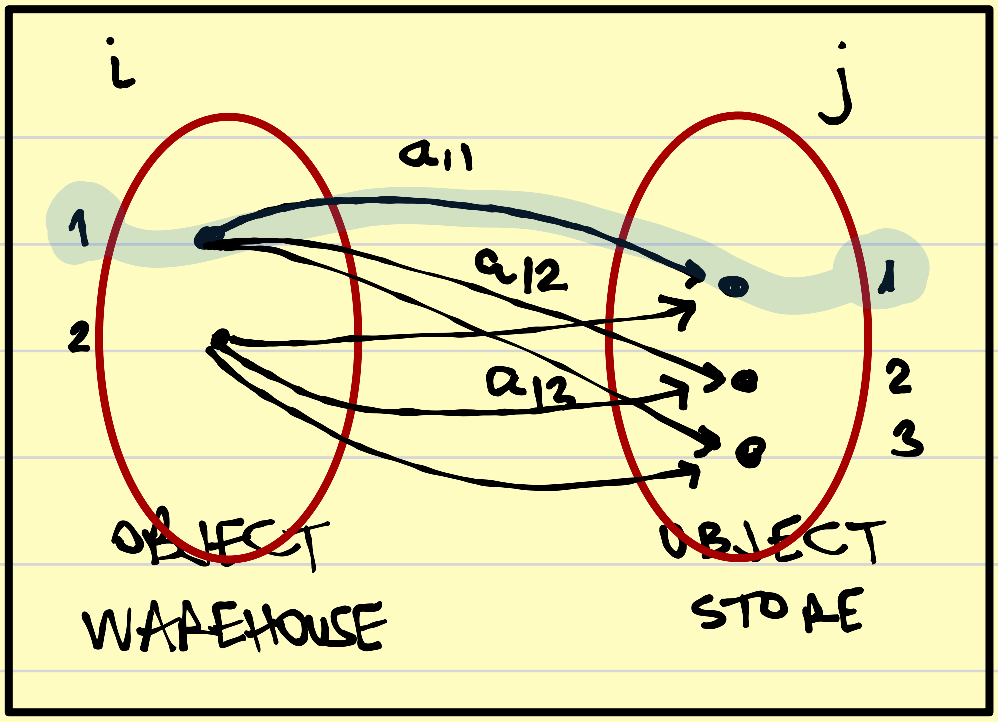
$\vec{d} = \begin{bmatrix} 5 \\ 6 \\ 3 \\ 7 \end{bmatrix}$ parts/week

MATRIX. A matrix is a rectangular array of numbers.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

m rows
n columns

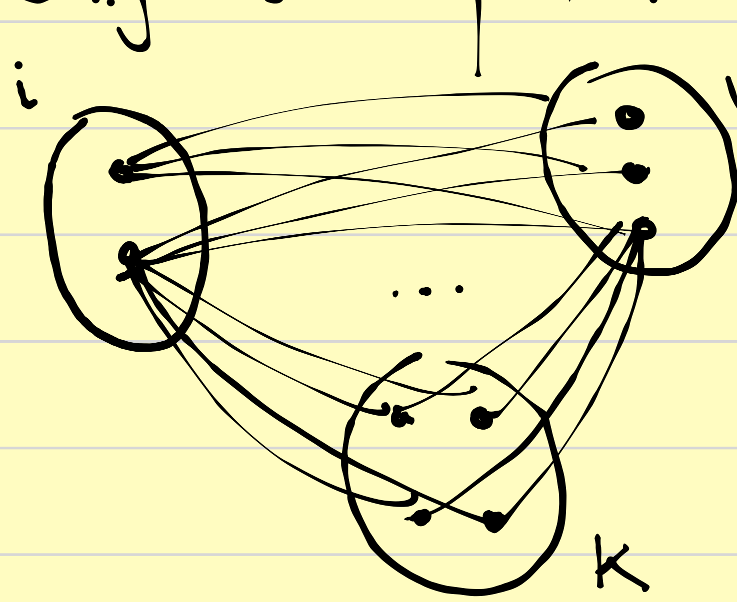
A matrix can represent a relation between two groups (indexed) of objects.

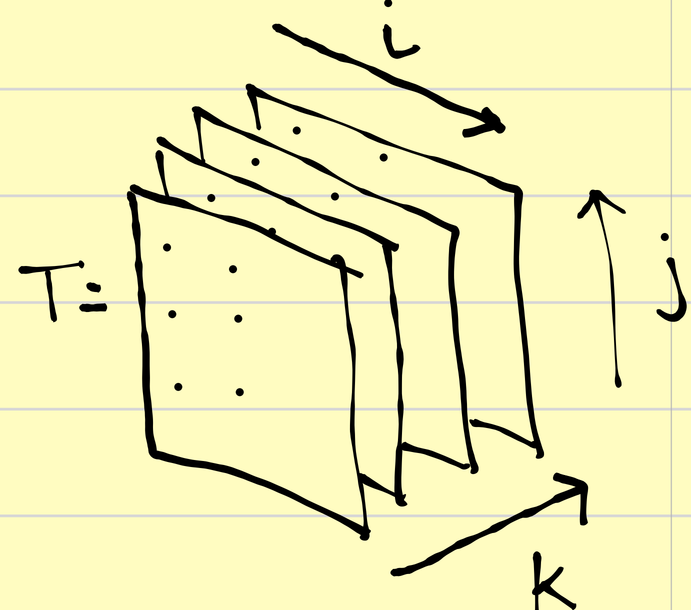


$$A = \underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}}_{\text{index store}} \Big\} \text{index WH}$$

TENSOR is a multi-index array. T_{ijk}

Example: quantity shipped from warehouse ..i.. to store ..j.. on day ..k..





- A matrix $A \in \mathbb{R}^{m \times n}$ defines a linear map $A : \mathbb{R}^n \to \mathbb{R}^m$.
- A matrix is not just a table, it is a rule that maps one vector of quantities into another vector of quantities.

Let $\vec{e_1}, \vec{e_2}, \dots, \vec{e_n}$ be the standard basis of $\mathbb{R}^n$, and $\vec{f_1}, \vec{f_2}, \dots, \vec{f_m}$ the standard basis of $\mathbb{R}^m$.

The j-th column of A is exactly: $A.e_j = \sum_{i=1}^{m} a_{ij} f_i$

So each entry a_{ij} tells us how the j-th source basis element contributes to the i-th target basis element.

> A matrix represents a weighted relation between two indexed groups of elements.

Vector operations and their logistics meaning.

If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ then:

Addition: $x + y = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix}$: combines effects componentwise

Scalar multiplication: $\alpha x = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \alpha x_3 \end{bmatrix}$: changes scale

Example: suppose the current stock at three stores is $s = \begin{bmatrix} 100 \\ 70 \\ 90 \end{bmatrix}$ and demand is $d = \begin{bmatrix} 120 \\ 80 \\ 100 \end{bmatrix}$.

Then the shortage vector $h = d - s = \begin{bmatrix} 120-100 \\ 80-70 \\ 100-90 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ 10 \end{bmatrix}$

Dot product: for two vectors $\vec{c}, \vec{x} \in \mathbb{R}^n$, the dot product is $\vec{c}^T \cdot \vec{x} = \sum_{i=1}^{n} c_i x_i$

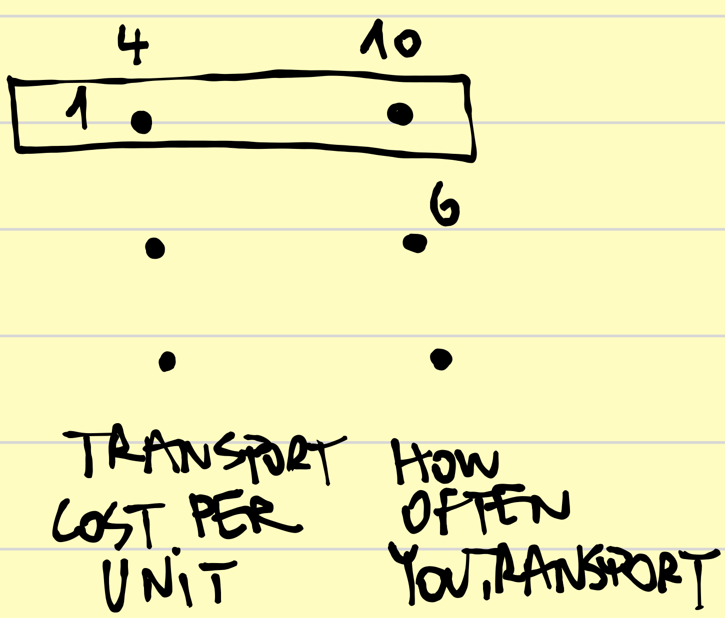
Logistics interpretation:

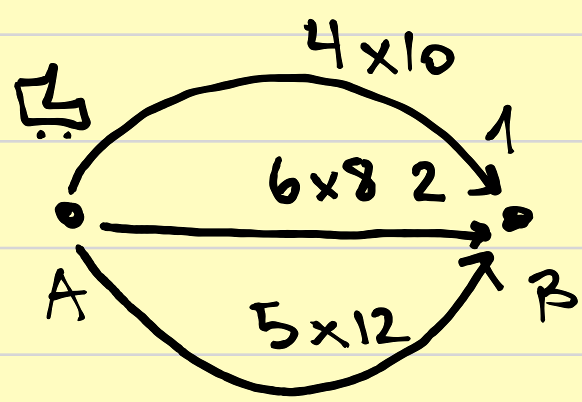
Let $c = \begin{bmatrix} 4 \\ 6 \\ 5 \end{bmatrix}$ be unit transport costs and $x = \begin{bmatrix} 10 \\ 8 \\ 12 \end{bmatrix}$ be shipment quantities.

What is the total cost?

$$\text{Cost} = \vec{c}^T . \vec{x} = \begin{bmatrix} 4 & 6 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \\ 12 \end{bmatrix} = 4.10 + 6.8 + 5.12 = 148$$

$$\boxed{\text{UNIT VECTOR COST}^T \times \text{QUANTITY VECTOR} = \text{TOTAL COST}}$$





$4 \times 10 + 6 \times 8 + 5 \times 12$

Linear combinations.

A vector $\vec{x} = \alpha_1 \vec{v_1} + \alpha_2 \vec{v_2} + \dots + \alpha_n \vec{v_n}$ is a linear combination of $\vec{v_1}, \vec{v_2}, \dots, \vec{v_n}$.

This means we build one plan, state, or demand profile from basic building blocks.

Example.

A company ships along three fixed routes.
The unit costs are:

$$\vec{c} = \begin{bmatrix} 3 \\ 7 \\ 5 \end{bmatrix} \quad \text{unit cost } \left[\text{cost}/\text{unit}\right]$$

The quantities are:

$$\vec{x} = \begin{bmatrix} 20 \\ 10 \\ 15 \end{bmatrix} \quad \text{quantities } [\text{unit}]$$

The total transport cost:

$$\vec{c}^{\,T} \cdot \vec{x} = 3 \cdot 20 + 7 \cdot 10 + 5 \cdot 15 = 210 \cdot \left[\frac{\text{cost}}{\cancel{\text{unit}}}\right] \cdot \left[\cancel{\text{unit}}\right]$$

Matrix-vector multiplication: one matrix acting on one vector.

$$\boxed{\begin{array}{l} \text{Let } A \in \mathbb{R}^{m \times n},\ x \in \mathbb{R}^n \text{ then } y = Ax \in \mathbb{R}^m \\ \text{and each component } \quad y_i = \sum_{j=1}^{n} a_{ij} \cdot x_j \end{array}}$$

A matrix transforms one vector $\vec{x} \in \mathbb{R}^n$ into another $\vec{y} \in \mathbb{R}^m$.

Dimensions must fit: $(m \times n) \cdot (n \times 1) = (m \times 1)$

$$A \cdot \vec{x} = \vec{y}$$

Expanded:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

Interpretation:
- each row of A forms one weighted sum of components of $\vec{x}$.
- each column of A tells how one input component influences all outputs.

Logistics example: supplier-to hub allocation

Suppose 3 suppliers provide quantities $\vec{q} = \begin{bmatrix} 50 \\ 40 \\ 60 \end{bmatrix}$ to two hubs.

The allocation matrix is:

$$A = \begin{array}{c|c} & \begin{array}{ccc} S1 & S2 & S3 \end{array} \\ \begin{array}{c} \text{HUB1} \\ \text{HUB2} \end{array} & \begin{bmatrix} 0.6 & 0.3 & 0.2 \\ 0.4 & 0.7 & 0.8 \end{bmatrix} \end{array}$$

Now comput $\vec{h} = A\vec{q}$

$$\vec{h} = \begin{bmatrix} 0.6 & 0.3 & 0.2 \\ 0.4 & 0.7 & 0.8 \end{bmatrix} \begin{bmatrix} 50 \\ 40 \\ 60 \end{bmatrix} = \begin{bmatrix} 0.6 \cdot 50 + 0.3 \cdot 40 + 0.2 \cdot 60 \\ 0.4 \cdot 50 + 0.7 \cdot 40 + 0.8 \cdot 60 \end{bmatrix}$$

$$\vec{h} = \begin{bmatrix} 54 \\ 96 \end{bmatrix}$$

← UNITS RECEIVED by HUB 1

← " " " " 2.